Worksheet: Evaluating LLMs and Code Models

CS 5090 · Software Engineering with Generative AI · Week 8 · collected for completion credit. Discussing with your neighbor is encouraged; write your own answer. No calculator needed; the numbers are chosen to work by hand.

Name: _____

Part 1 · Metrics for open-ended tasks, and for code

match **1.1** There are four ways to get a metric for an open-ended task, plus a fifth that only code has. Write the letter of the approach next to each example.

(A) convert to a closed-ended task (B) reference-based heuristic (C) humans (D) LLM as judge (E) execution

Example	Approach
MMLU: 57-subject multiple-choice exam, auto-graded	
BLEU score against one reference translation	
Chatbot Arena: the public votes between two anonymous answers	
AlpacaEval: GPT-4 compares each answer with a baseline answer	
HumanEval: the generated function is run against hidden unit tests	

think **1.2** The reference solution is `return sum(xs)`. Two model outputs:

Output 1: `t = 0; for x in xs: t += x; return t` Output 2: `return sum(xs[1:])`

Which output is **correct**? _____ Which has the **higher BLEU** (more tokens in common with the reference)? _____

In one sentence: what does this say about using BLEU for code?

work it out **1.3** pass@k with the unbiased estimator. You draw **n = 5** samples for one problem and **c = 2** of them pass the tests.

pass@k = 1 − C(n − c, k) / C(n, k) · pass^k = C(c, k) / C(n, k) · C(3,1) = 3, C(5,1) = 5, C(3,2) = 3, C(5,2) = 10, C(2,2) = 1

pass@1 = _____ pass@2 = _____ pass^2 (both of 2 attempts pass) = _____

A coding assistant that gives the user one answer: which of the three numbers describes what the user experiences? _____

recall **1.4** Two papers both report "pass@1 = 70%" on HumanEval. Name **two** things about the setup that, if different, make the two numbers not comparable.

Part 2 · Perplexity, leaderboards, Goodhart's law

work it out **2.1** A model's average per-token loss (negative log-likelihood) on a test set is 3.0, and $e^{3.0} \approx 20$. Its perplexity is _____. In words, what does that number mean about the model's next-token prediction?

recall **2.2** True or false, circle one each:

T / F Perplexity numbers from two models with different tokenizers and test sets can be compared directly.

T / F Chatbot Arena measures which answer people *prefer*, not which answer is correct.

T / F AlpacaEval needs a stronger "oracle" model to act as the judge.

explain **2.3** AlpacaEval's judge favours **longer** answers: prompting the same model to be verbose instead of concise raised its win rate from 22.9% to 64.3%. (a) What did models that were tuned for the leaderboard start doing? (b) What was the fix? (c) Write Goodhart's law in your own words.

Part 3 · Benchmarks and contamination

match **3.1** Write the benchmark name next to its description: **HumanEval**, **HumanEval+**, **SWE-bench Verified**, **LiveCodeBench**.

Description	Benchmark
Real GitHub issues; the fix is judged by the repository's own tests	

164 hand-written Python functions with a few tests each	
Contest problems collected over time, each tagged with its release date	
The same 164 problems with many more generated tests	

interpret **3.2** GPT-4's pass@1 drops from 88.4 on HumanEval to 76.2 on HumanEval+. What does the drop mean? Circle one.
(a) GPT-4 got worse between the two runs (b) the original tests let some wrong solutions pass (c) HumanEval+ has harder problems (d) HumanEval+ is contaminated

match **3.3** Three ways to detect contamination. Write **time split**, **perturbation**, or **likelihood** next to each:

Method	Name
Compare accuracy on problems released before vs. after the model's training cutoff	
Check whether the model assigns suspiciously high probability to the exact test text	
Rewrite the problems with an LLM and see whether the score drops	

recall **3.4** You are reading a leaderboard's "about" page. Write the three questions from the lecture that tell you whether to trust its numbers.

Part 4 · Judges and new capabilities

match **4.1** Each LLM-judge bias has a fix. Draw a line (or write the letter):

Bias	Fix
Position (prefers the first answer)	(a) control for length, or report a length-controlled win rate
Verbosity (prefers the longer answer)	(b) use a judge from a different model family
Self-preference (prefers its own style)	(c) run both orders and average

decide **4.2** Your Project 2 uses GPT-4 as a judge. You label 100 outputs by hand: the judge agrees with you on 55 of them. In the MT-Bench paper, two humans agreed with each other about 81% of the time. Can you report the judge's numbers as-is? What two things should you do first?

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apply **4.3** HumanEvalComm measures whether a code model **asks a clarifying question** when the problem is ambiguous. (a) Why can pass@k alone not measure this? (b) For each of its metrics, write which approach from 1.1 it uses (A–E):
Communication Rate (how often the model asks) ____ Good Question Rate (is the question useful?) ____ Pass@1 after the simulated user answers ____

interpret **4.4** In GrowthHacker, agents edit code to lower an estimation error. One framework reports an **average** improvement of 37.9% but a **median** of 1.6%. What does the gap tell you about its runs? Why does the benchmark flag any change above 9,999% as EXTREME instead of counting it as a win?

Part 5 · Testing with LLMs

think **5.1** Your function has an off-by-one bug. You paste it into an assistant and ask for unit tests. The tests pass. What went wrong, and where must the oracle (the expected values) come from instead?

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work it out **5.2** A mutation tool makes 20 mutants of your module; your test suite fails on 14 of them. Mutation score = _____%. A different suite runs every line (100% coverage) but has no assertions: its coverage is _____% and its mutation score is _____%. Which number tells you more about the tests? _____